

Multimedia

Vorlesung

Hannes Kaufmann

Virtual & Augmented Reality Group

Research Unit Computer Graphics

Institute of Visual Computing & Human-Centered Technology

Based on material by Prof. Christian Breiteneder

Psychoacoustics

- Scientific study of sound perception and audiology: How humans perceive various sounds
- **Study on your own** the Youtube course “Psychoakustik” (German)
https://www.youtube.com/watch?v=8_TxY93L8xE&list=PLftcEygAp_6lk8SpKNesLim_omxfXMbmo&index=1
11 parts, ~ 107 min (total)

Inverted Classroom: Psychoacoustics Questions

- How do we hear?
- What happens inside the cochlea?
- What are the different ways to judge the location & direction of sound?
- What is 1 MEL? What is 1 BARK?
- What is critical bandwidth?
- How large is a critical band?
- What is the connection between critical bands and hair cells?

Inverted Classroom: Psychoacoustics Questions

- Explain and annotate the figure ->
- What is masking?
- What is temporal masking?
- Describe a psychoacoustic phenomenon (e.g. Shepard Scala, Tritonus Paradoxon,...)

